Testy dobrej zhody

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```
library(latexpdf)
```

Pearsonov χ^2 test dobrej zhody

Overujeme zhodu s diskretnym rozdelenim

Priklad: Hadzeme kockou 600x. Vysledky su v datovom subore pocetnost. Overte, ci je kocka homogenna. Statisticky: overte zhodu s rovnomernym diskretnym rozdelenim {1,2,3,4,5,6}. Rozdelenie do tried je prirodzene, 1- 6 a padalo to takto

```
pocetnost <- c(122,61,98,115,79,125)
```

Data musia dlzky 100

```
data <- rep(1:6,pocetnost)
```

Do prikazu pre χ^2 idu aj teoreticke pocetnosti, ktore musi pouzivatel zratat, vynimku tvori rovnomerne rozdelenie, vtedy nemusime nic zadat.

```
chisq.test(pocetnost)
```

```
##
##  Chi-squared test for given probabilities
##
## data:  pocetnost
## X-squared = 33, df = 5, p-value = 3.763e-06
```

P hodnota<0.05, teda zamietame zhodu s rovnomernym rozdelenim. Kocka nie je homogenna.

Pocas 100 dni sme sledovali pocty roztrhnutia vlakna na pletacom stroji po generalnej oprave. Overte zhodu s Poissonovym rozdelenim s parametrom $\lambda = 3$, t.j. v priemere tri roztrhnutia za den 0 roztrhnuti 4x, 1 roztrhnutie 10x..., namerali sme 0-8 roztrhnuti, Poissonovo rozdelenie teoreticky nadobuda hodnoty az do nekonecna

```
d1 <- c(4,10,26,27,13,10,4,4,2)
```

Vypocet teoretickych pocetnosti

```
pp <- dpois(0:7,3)
pp
```

```
## [1] 0.04978707 0.14936121 0.22404181 0.22404181 0.16803136 0.10081881 0.05
040941
## [8] 0.02160403
```

```r
sum(pp)
```

```
## [1] 0.9880955
```

Posledna pravdepodobnost je pre triedu 8 a viac roztrhnuti, aby to sedelo, ze ista udalost je 1

```r
pp[9] <- 1-sum(pp)
pp
```

```
## [1] 0.04978707 0.14936121 0.22404181 0.22404181 0.16803136 0.10081881 0.05
040941
## [8] 0.02160403 0.01190450
```

```r
chisq.test(d1,p=pp)
```

```
## Warning in chisq.test(d1, p = pp): Chi-squared approximation may be incorr
ect
```

```
##
##  Chi-squared test for given probabilities
##
## data:  d1
## X-squared = 6.5369, df = 8, p-value = 0.5873
```

Upozornujuca sprava hovori, ze nie je splneny predpoklad teoreticke pocetnosti v triedach>=5. Za takejto sitacie Phodnota>0.05, nezamietame H_0. Mozmeme dat prikaz v tvare

```r
chisq.test(d1,p=pp,simulate.p.value = T)
```

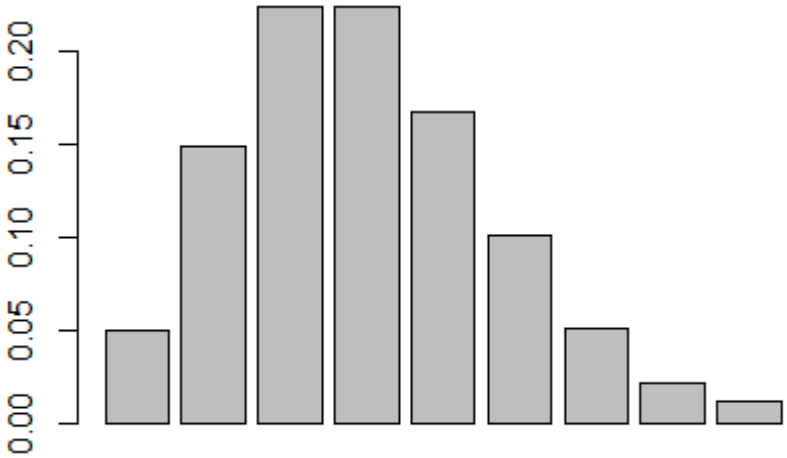
```
##
##  Chi-squared test for given probabilities with simulated p-value (based
##  on 2000 replicates)
##
## data:  d1
## X-squared = 6.5369, df = NA, p-value = 0.5692
```

vysledok je rovnaky ako hore.

Grafy pre GoF

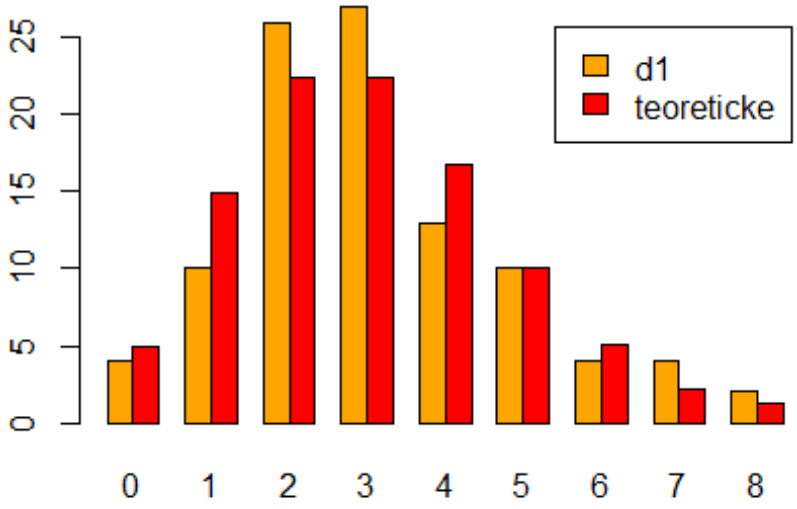
graf pre predosly priklad

```r
hustota <- pp
x <- 0:8
tabulka <- data.frame(hodnota=x,pravdepodbnost=pp)
barplot(tabulka$pravdepodbnost)
```



teoreticke pocetnosti

```
teoreticke <- 100*pp
pocetnosti <- rbind(d1,teoreticke)
barplot(pocetnosti,beside=T,names.arg=x,legend=T,
        col=c("orange","red"))
```



Kniznica EnvStats

Trochu komplikovanejsi tvar prikazu, ale rovno dostaneme diagnosticke grafy prikazom plot.

```
library(EnvStats)
```

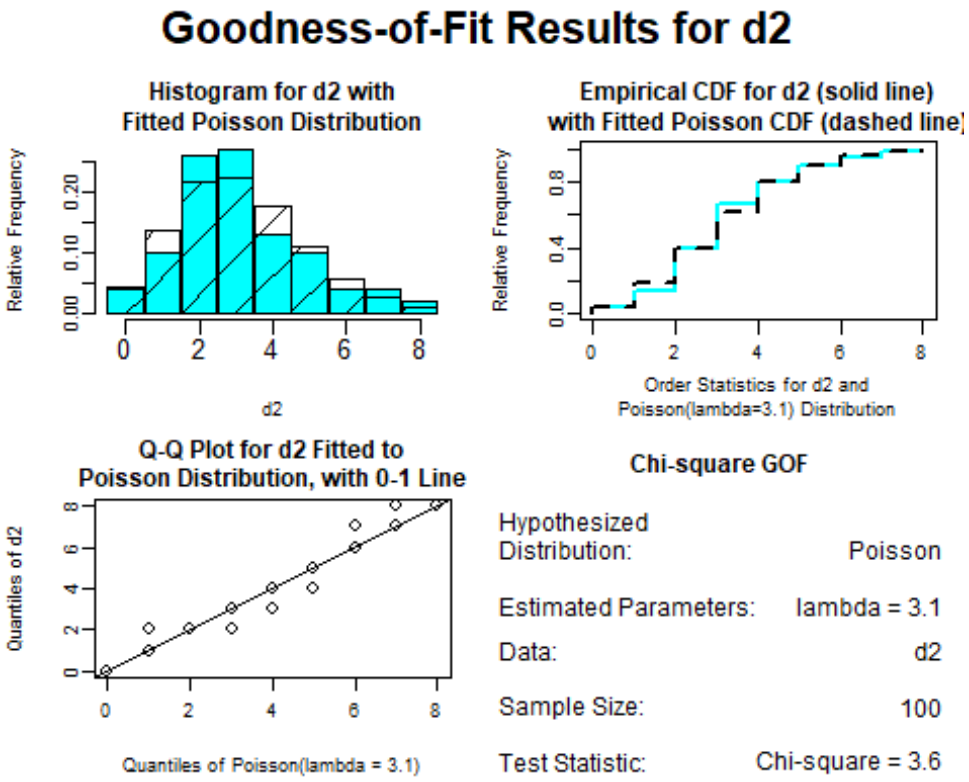
```
##
## Attaching package: 'EnvStats'
```

```
## The following objects are masked from 'package:stats':
##
##     predict, predict.lm
```

```
## The following object is masked from 'package:base':
##
##     print.default
```

data musime dat do tvaru ako boli merane

```
d2 <- rep(0:8,d1)
a <- gofTest(d2,test="chisq",dist="pois",
             cut.points=c(-1,1,2,3,Inf))
plot(a)
```



V zadani bolo, ze $\lambda = 3$

```
a <- gofTest(d2,test="chisq",dist="pois",
             param.list = list(lambda=3),
             cut.points=c(-1,1,2,3,Inf))
a$p.value

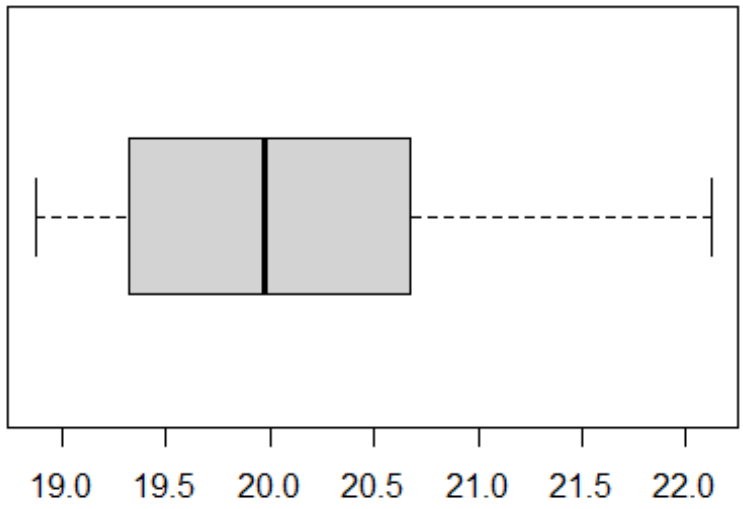
## [1] 0.3308123
```

p hodnota je>0.05, nezamietame H_0 zhoda s Poissonovym rozdelenim

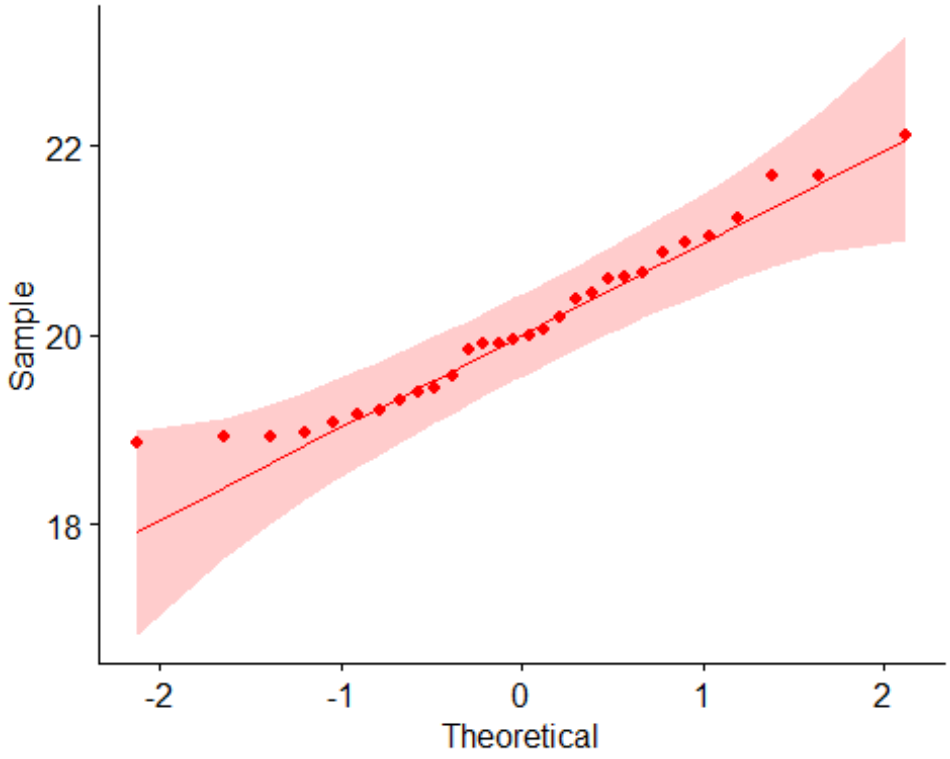
KS test dobrej zhody

Testujeme meraci pristroj, 100x opakujeme meranie rozmeru suciastky, ktora ma spravny rozmer (stredna hodnota) 20.1. Pristroj ma smerodajnu odchylku 0.9. KS testom testujte hypotezu, ze data su normalne rozdelene s danymi parametrami.

```
d1 <- c(19.32, 19.57, 19.08, 20.87, 19.9, 18.92, 20.98, 20.18,
        20.67, 18.87, 21.69, 19.9, 19.16, 20.44, 21.68, 21.23,
        19.44, 19.95, 20.59, 20.62, 21.05, 20,18.96, 22.12, 19.39,
        19.85, 20.39, 18.93, 20.06,19.2)
boxplot(d1,horizontal=T)
```




```
library(ggplot2)
library(ggpubr)
ggqqplot(d1,col="red")
```



Vidime aj pas spolahlivosti pre priamku, data su v pase -OK

```
b <- gofTest(d1,test="ks",
             param.list=list(mean=20.1,sd=0.9))

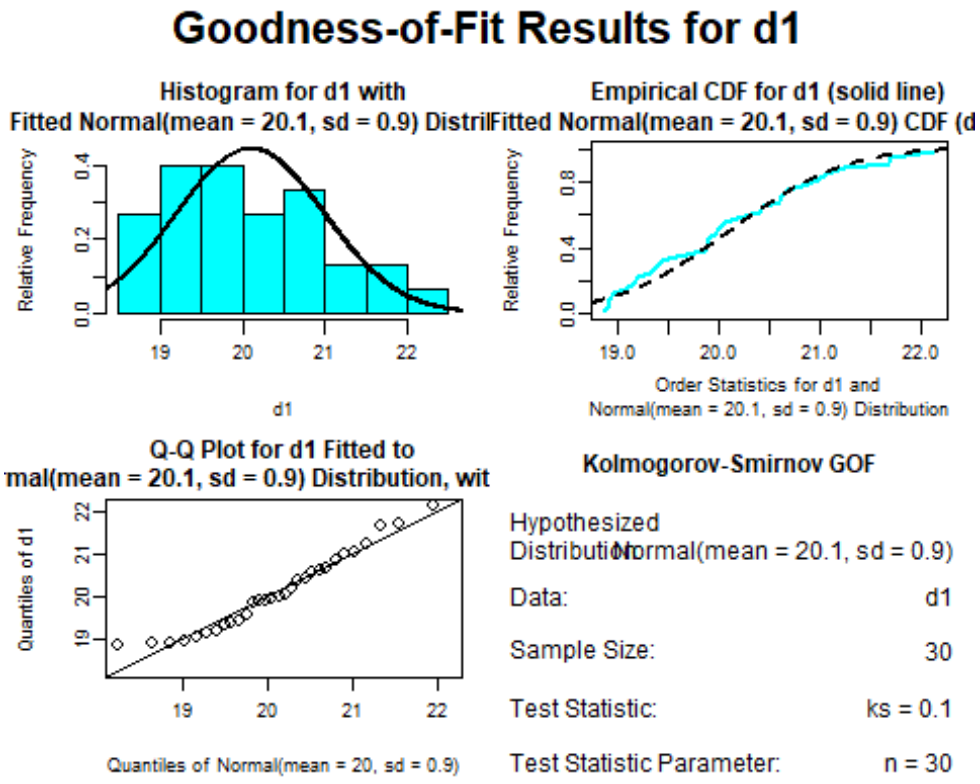
## Warning in ks.test.default(x = c(19.32, 19.57, 19.08, 20.87, 19.9, 18.92, :
## ties should not be present for the one-sample Kolmogorov-Smirnov test

b$p.value

## [1] 0.9158337
```

Nezamietame H_0.

```
plot(b)
```



Test nemozeme pouzit, ked nepozname parametre a parametre odhadujeme z dat. Vtedy modfikovany Lillieforsov test

```
mean(d1)

## [1] 20.10033

sd(d1)

## [1] 0.9098408
```

Modifikovany KS test Lillieforsov test. Kniznica nortest.

```
library(nortest)
lillie.test(d1)
```

```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  d1
## D = 0.099343, p-value = 0.6322
```

Neamietame H_0

KS test a zhoda s exponencialnym rozdelenim

Zivotnost vyrobkov ma exponencialne rozdelenie so strednou hodnotou 2000 dni. Exp ma parameter λ, kde plati$\lambda = \frac{1}{E(X)}$. Nas parameter je $. Pre data overte zhodu s danym rozdelenim.

```
zivotnost<-c(4996,976,294,7302,2420,1006,7654,2973,320,3169,3994,
             3989,2689,2215,391,4966,1959,9167,5365,2377)
ks.test(zivotnost,"pexp",1/2000)
```

```
##
##  Exact one-sample Kolmogorov-Smirnov test
##
## data:  zivotnost
## D = 0.3745, p-value = 0.005002
## alternative hypothesis: two-sided
```

Zamietame hypotezu o zhode s danym rozdelenim. Ale ked spocitame strednu hodnotu, vidime, ze asi kvoli zle urcenemu parametru.

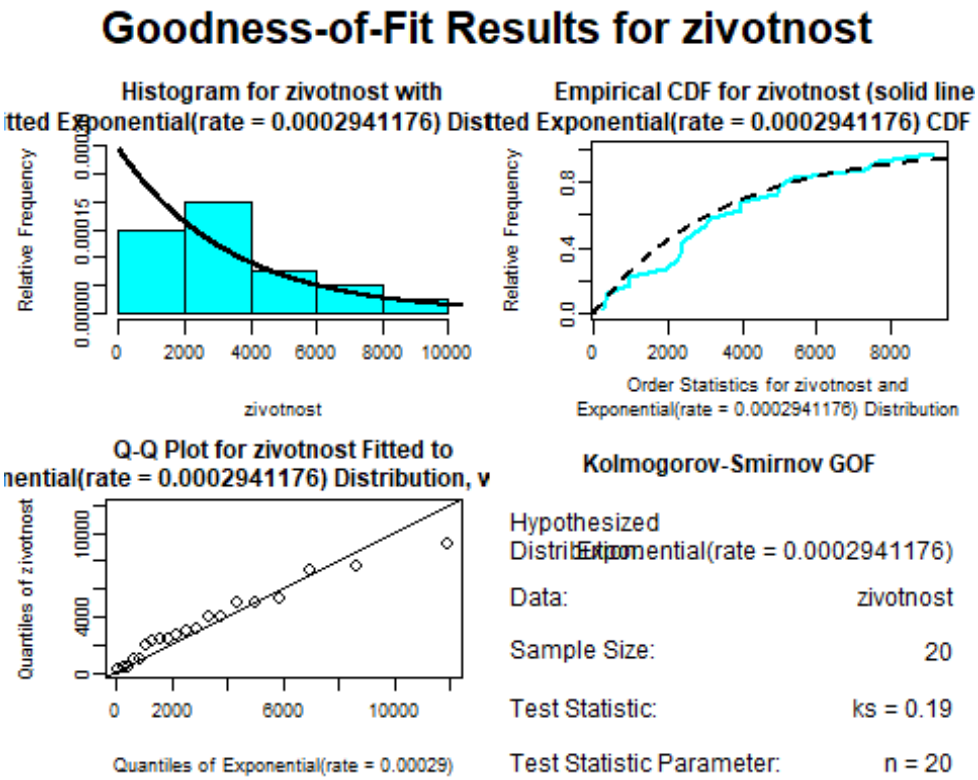
```
mean(zivotnost)
```

```
## [1] 3411.1
```

```
c <- gofTest(zivotnost,test="ks",dist="exp",
             param.list =list(rate=1/3400))
c$p.value
```

```
## [1] 0.4273924
```

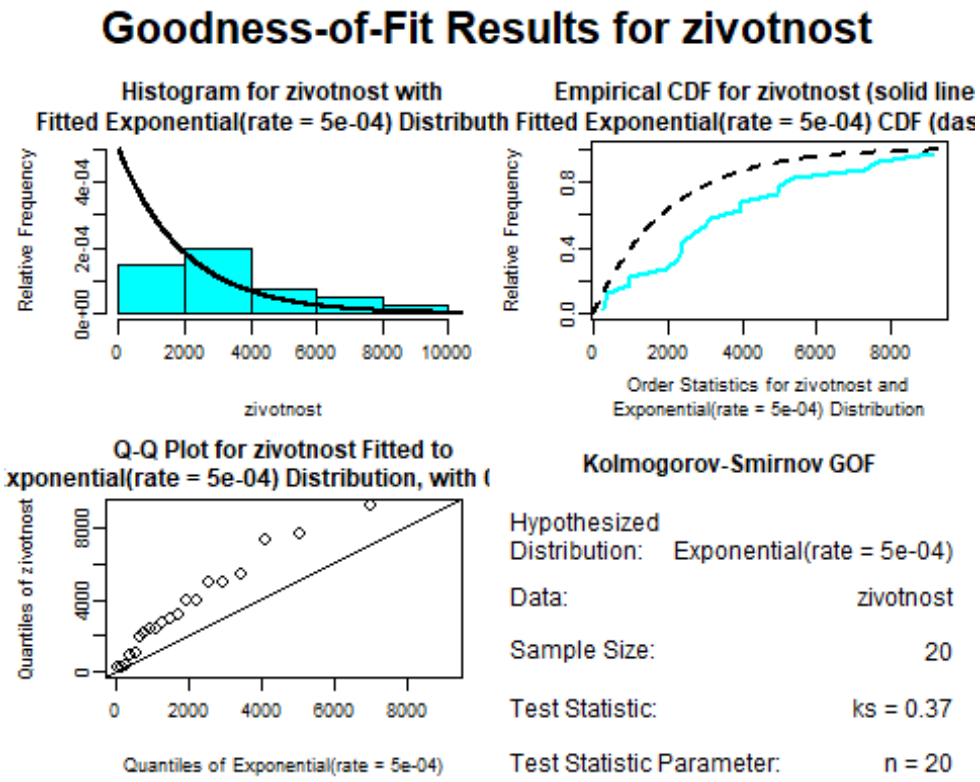
```
plot(c)
```




```
c <- gofTest(zivotnost,test="ks",dist="exp",
             param.list =list(rate=1/2000))
c$p.value

## [1] 0.005001848

plot(c)
```

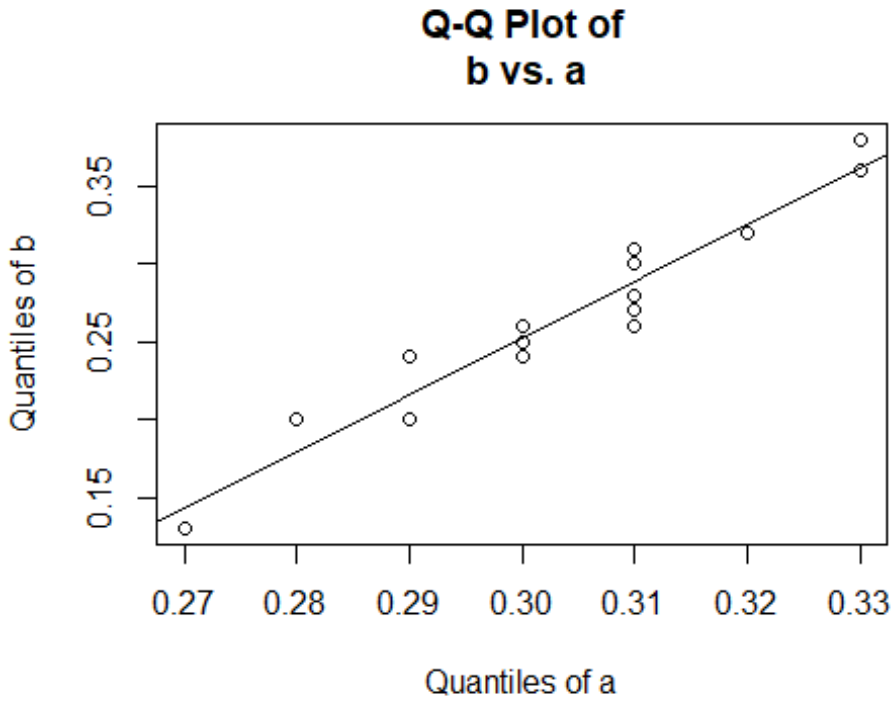


Dvojvyberove testy dobrej zhody

Testujeme dva vybery, ze maju rovnake rozdelenie. kedze rozdelenie nie je specifikovane, tak dvojvyberovy KS test patri k neparametrickym metodam.

Komercne laboratorium testuje pritomnost protilatok na danu chorobu metodami A, B. Davaju rovnake vysledky?

```
a <- c(0.28, 0.31, 0.27, 0.33, 0.33, 0.29, 0.29, 0.3, 0.31, 0.31, 0.32,
       0.31, 0.3, 0.31, 0.3)
b <- c(0.24, 0.25, 0.28, 0.27, 0.2, 0.32, 0.13, 0.26, 0.38, 0.24, 0.36,
       0.31, 0.3, 0.2, 0.26)
qqPlot(a,b,add.line=T)
```




```
ks.test(a,b)
```

```
##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  a and b
## D = 0.53333, p-value = 0.02087
## alternative hypothesis: two-sided
```

p hodnota <0.05, teda nedavaju rovnake vysledky.

Testy urcene pre normalne rozdelenie

Znamy Shapiro Wilk test, ma velku silu, parametre nemusime poznat, data d1 meraci pristroj

```
shapiro.test(d1)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  d1
## W = 0.95003, p-value = 0.1693
```

pouzitie aj Jarque Bera a Anderson Darling. Prvy test je v kniznici tseries.

```
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
jarque.bera.test(d1)
```

```
##
##  Jarque Bera Test
##
## data:  d1
## X-squared = 1.6204, df = 2, p-value = 0.4448
```

```
ad.test(d1)
```

```
##
##  Anderson-Darling normality test
##
## data:  d1
## A = 0.39641, p-value = 0.3485
```